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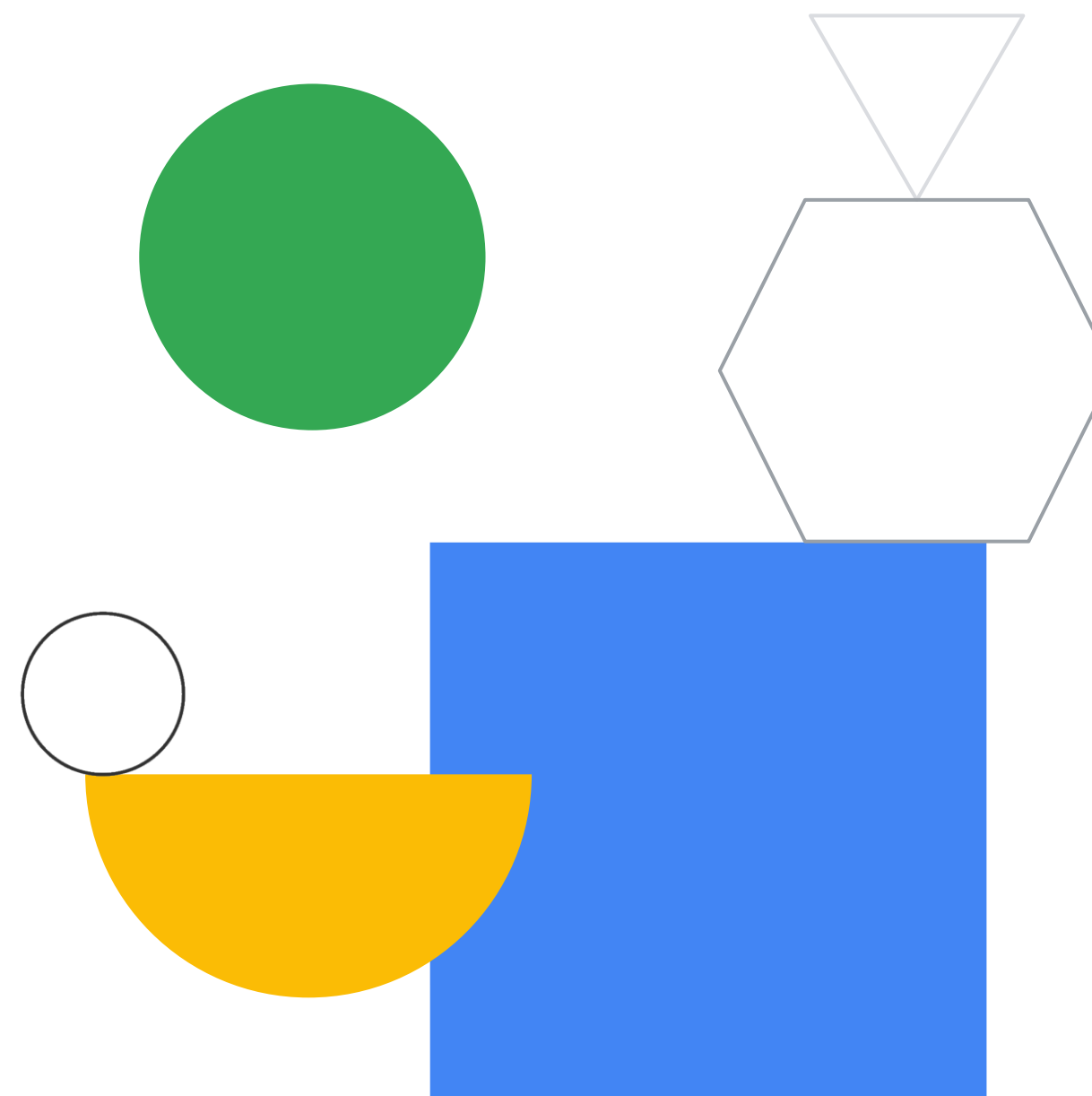
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Where do I store this stuff?

Module 4 - Storage
Google Cloud Computing Foundations



Course map

01 —

So, what's the cloud anyway?

02 —

Start with a solid platform

03 —

Use Google Cloud to build your apps

04 —

Where do I store this stuff?

05 —

There's an API for that!

06 —

You can't secure the cloud, right?

07 —

It helps to network

08 —

Keeping an eye on things

09 —

You have the data, but what are you doing with it?

10 —

Let machines do the work

Google Cloud skill badges

Objectives

- 01 Explore the different storage options available in Google Cloud.
- 02 Learn to differentiate between structured and unstructured storage in the cloud.
- 03 Examine how you can use Cloud Storage for unstructured data storage.
- 04 Explore the use case for relational vs NoSQL storage options, and identify the options available with Google Cloud.



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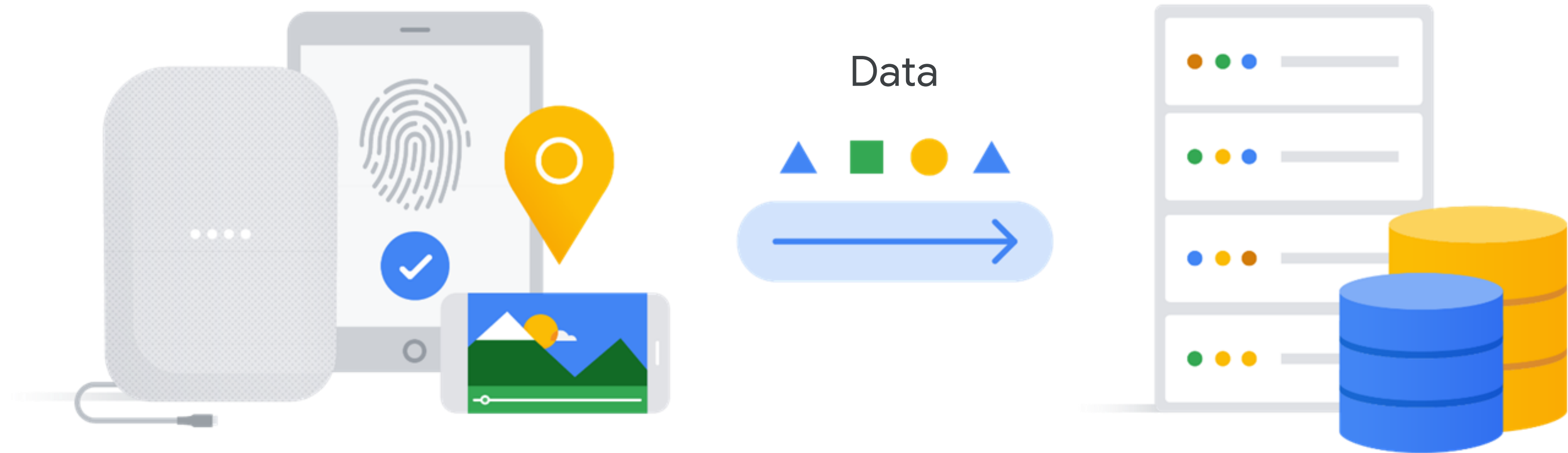
11 Cloud Bigtable as a NoSQL option

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Application needs to store data



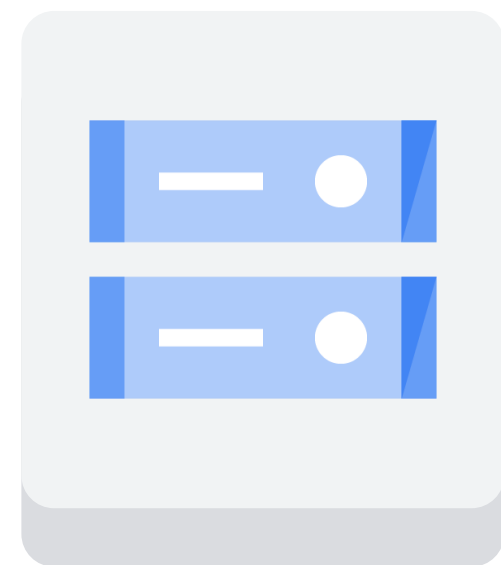
Storage offerings



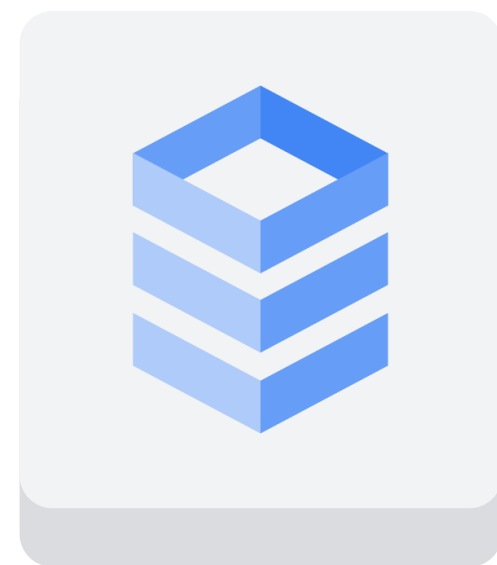
Google Cloud

- ✓ Relational databases
- ✓ Non-relational databases
- ✓ Worldwide object storage

Google Cloud database and storage services

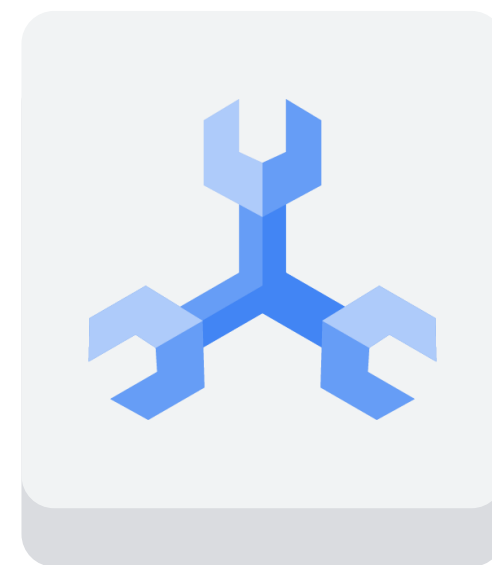


Cloud Storage



Cloud SQL

Relational
databases



Cloud Spanner

NoSQL databases

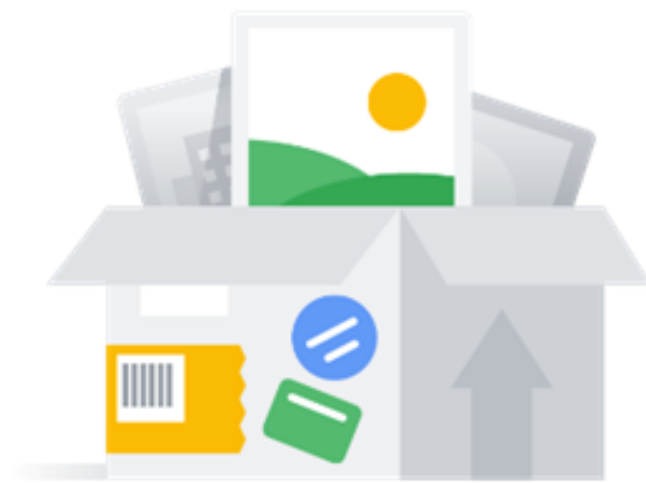


Firestore

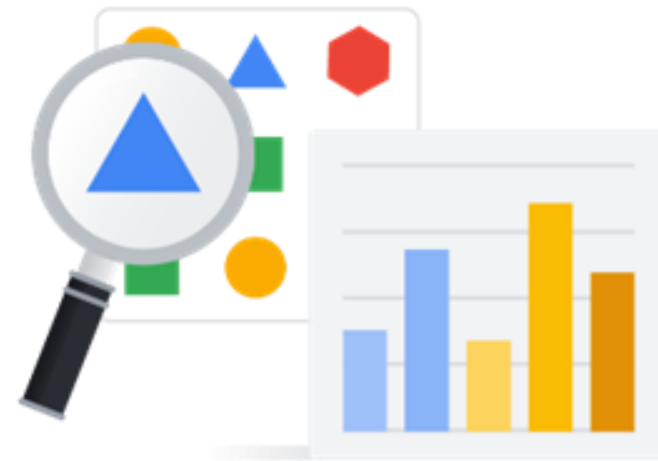


Cloud Bigtable

Common cloud storage use cases



Content storage
and delivery



Data analytics and
general compute

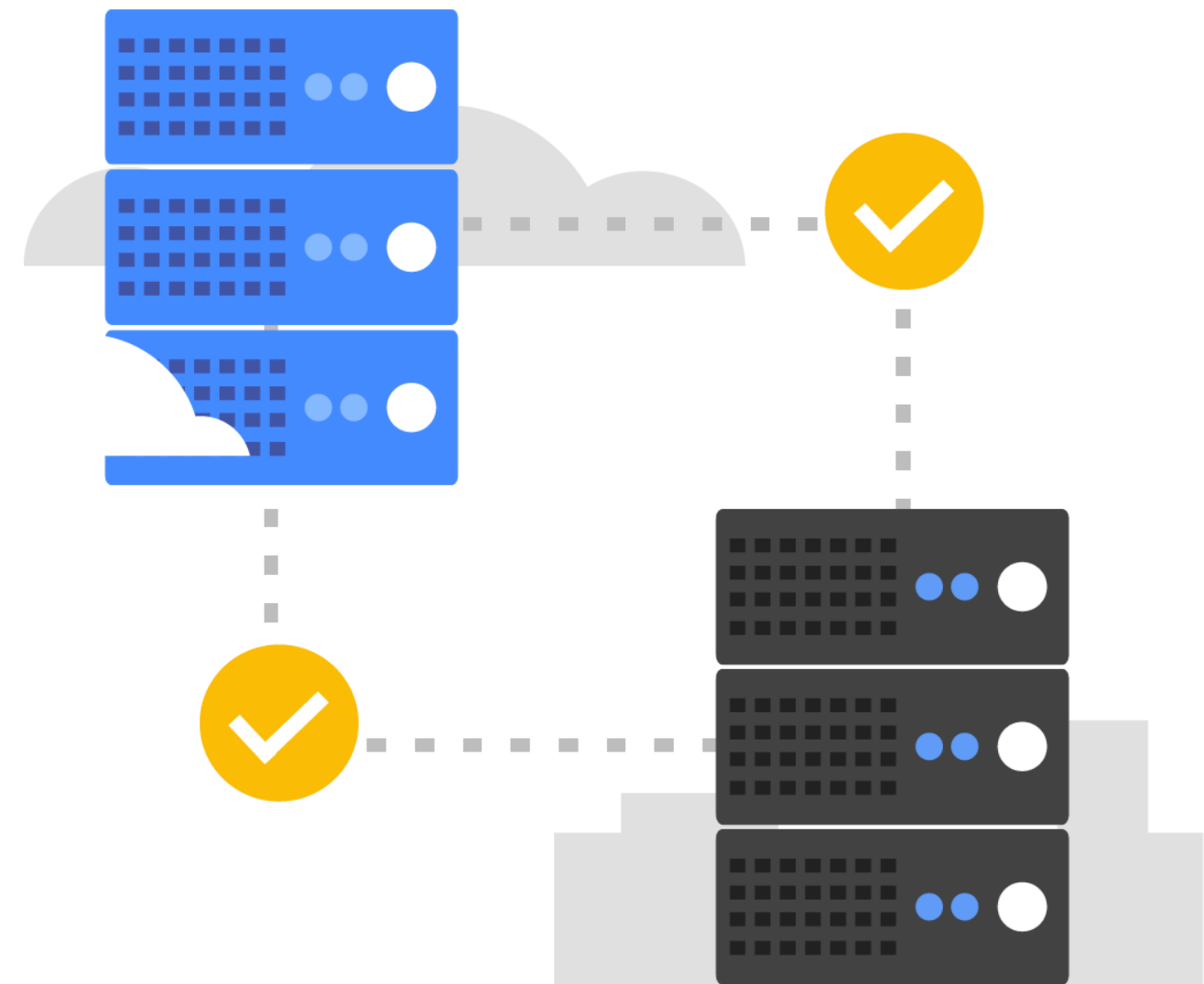


Backup and
archival storage

For users with databases, Google has two priorities

01 Migrate existing databases to the cloud, and move them to the right service.

02 Innovate, build or rebuild for the cloud, offer mobile applications, and plan for future growth



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Unstructured data → Cloud Storage

Unstructured



- ◆ Documents
- ▲ Images
- Audio files



Cloud
Storage

Structured data



Choose the best storage option

Structured



Transactional workload

SQL

Local/regional scalability



Cloud SQL

Global scalability



Cloud Spanner

NoSQL



Firestore

Analytical workload

SQL



BigQuery

NoSQL



Cloud Bigtable

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Cloud Storage is a fully managed scalable service



Website content

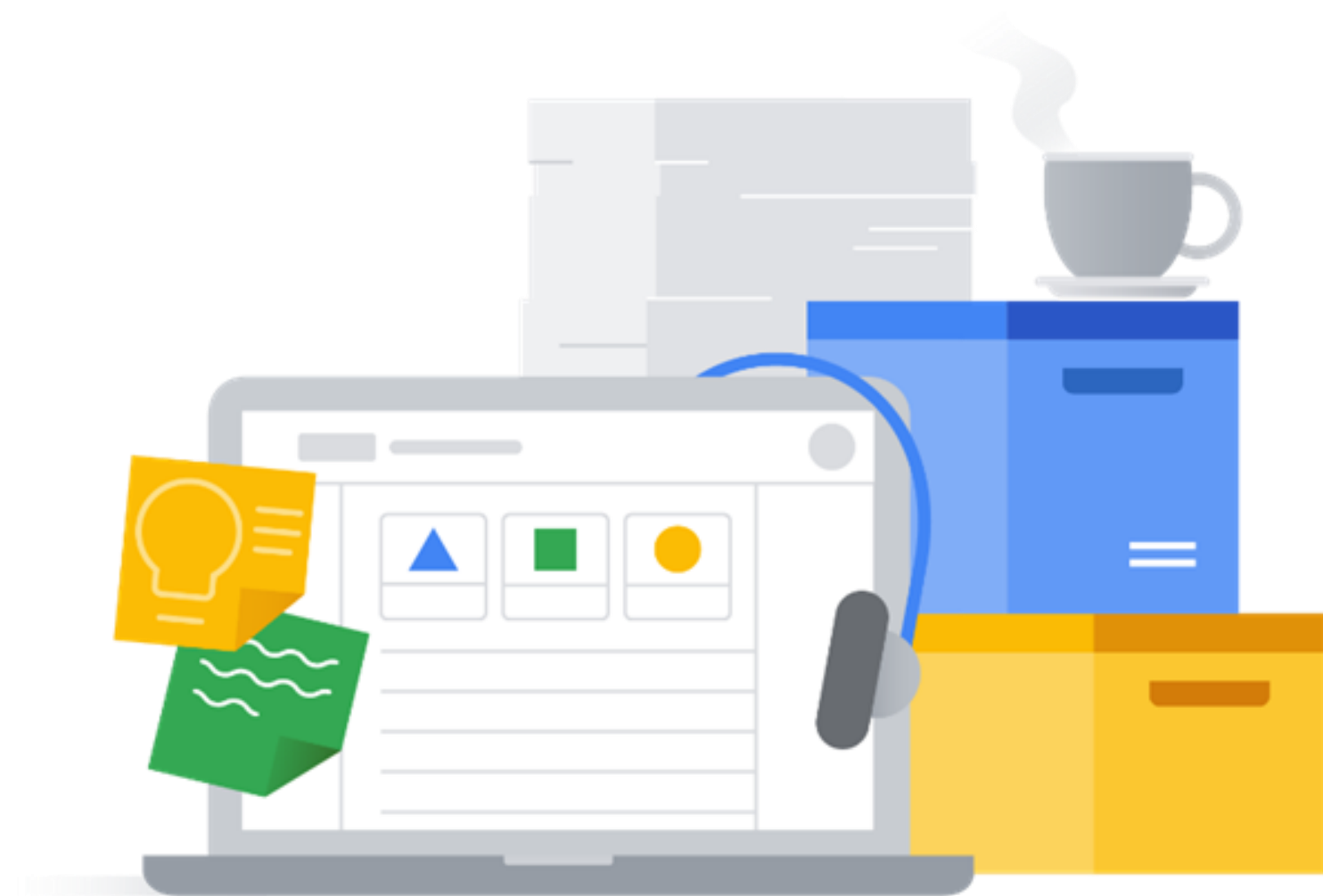
Archival & disaster recovery

Direct Download

Cloud Storage's common object storage use cases

Binary large-object (BLOB) storage

- ✓ Online content
- ✓ Backup and archiving
- ✓ Storage of intermediate results



Types of file storage



File storage



Block storage



Object storage

There are four basic Cloud Storage classes

Standard Storage



Hot data

Nearline Storage



Once per month

Coldline Storage



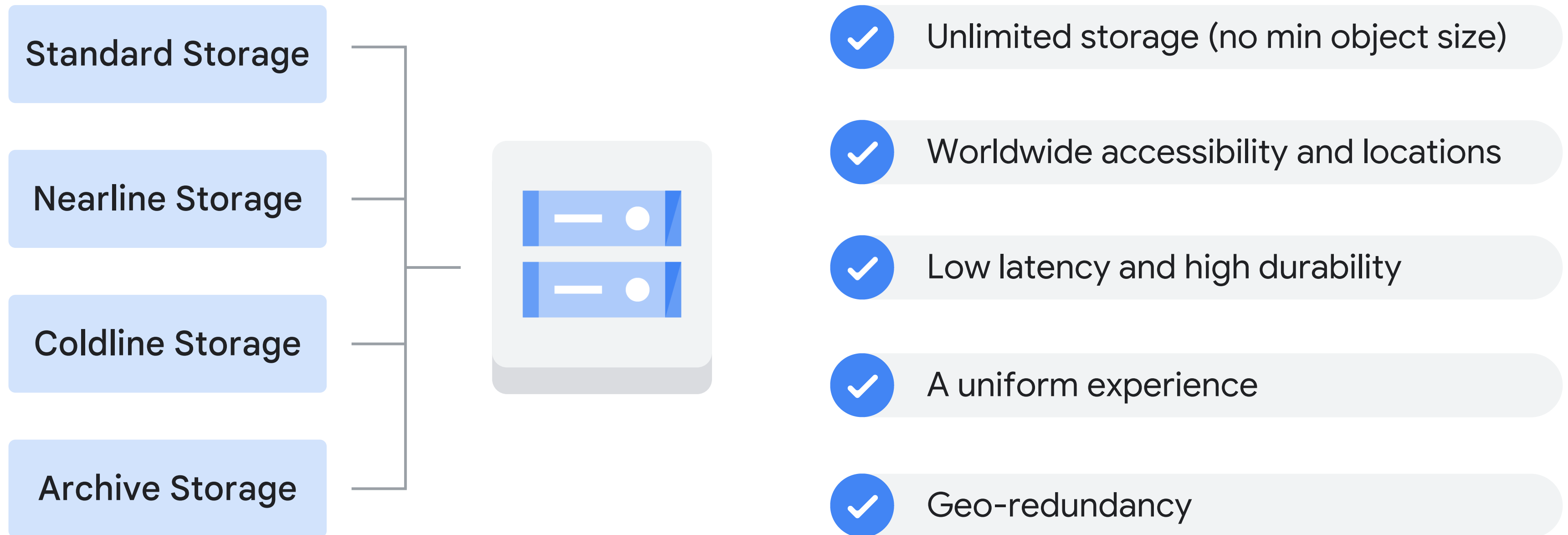
Once every 90 days

Archive Storage

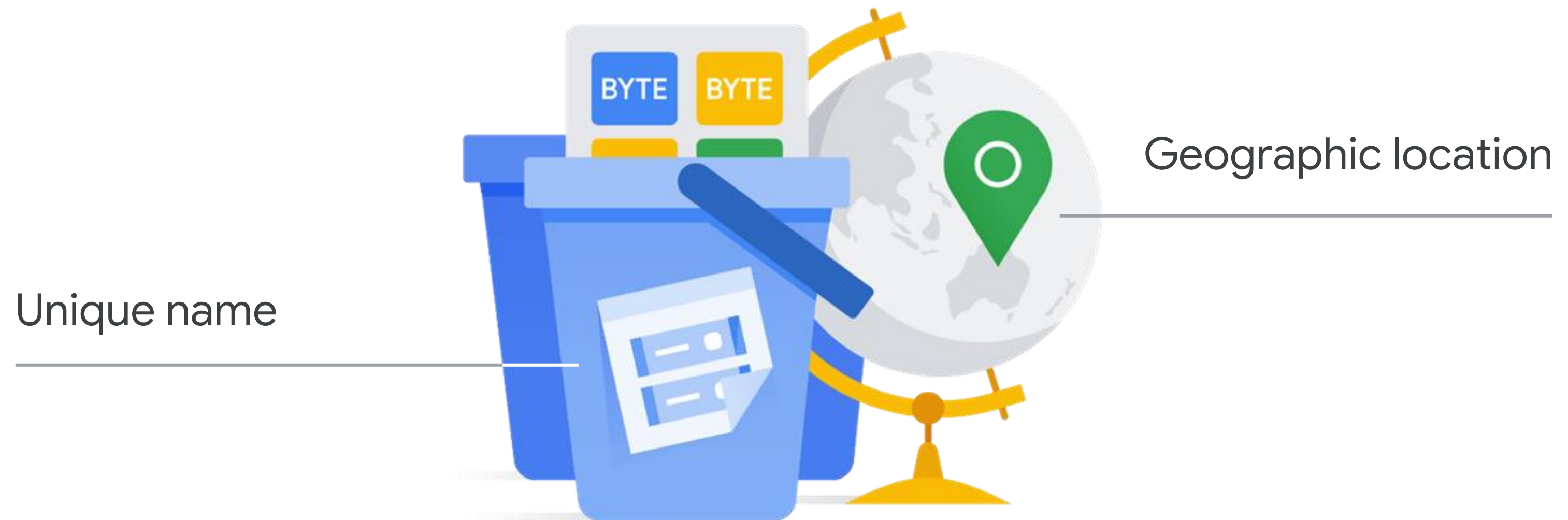


Once a year

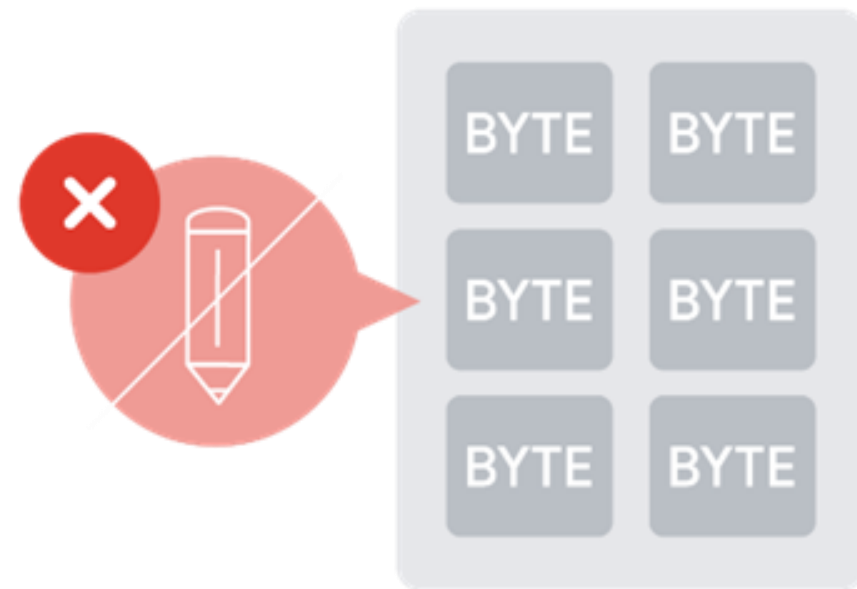
Each storage class has some common features



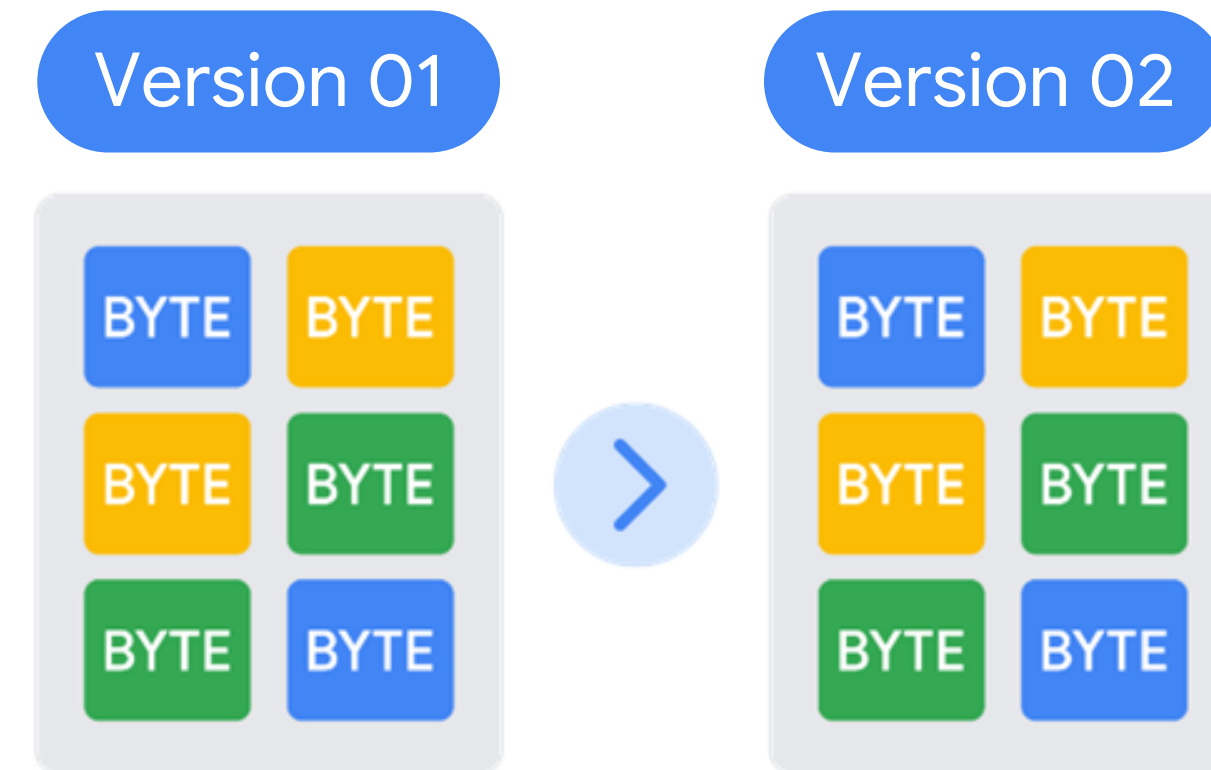
Cloud Storage files are organized into buckets



Cloud Storage objects are immutable

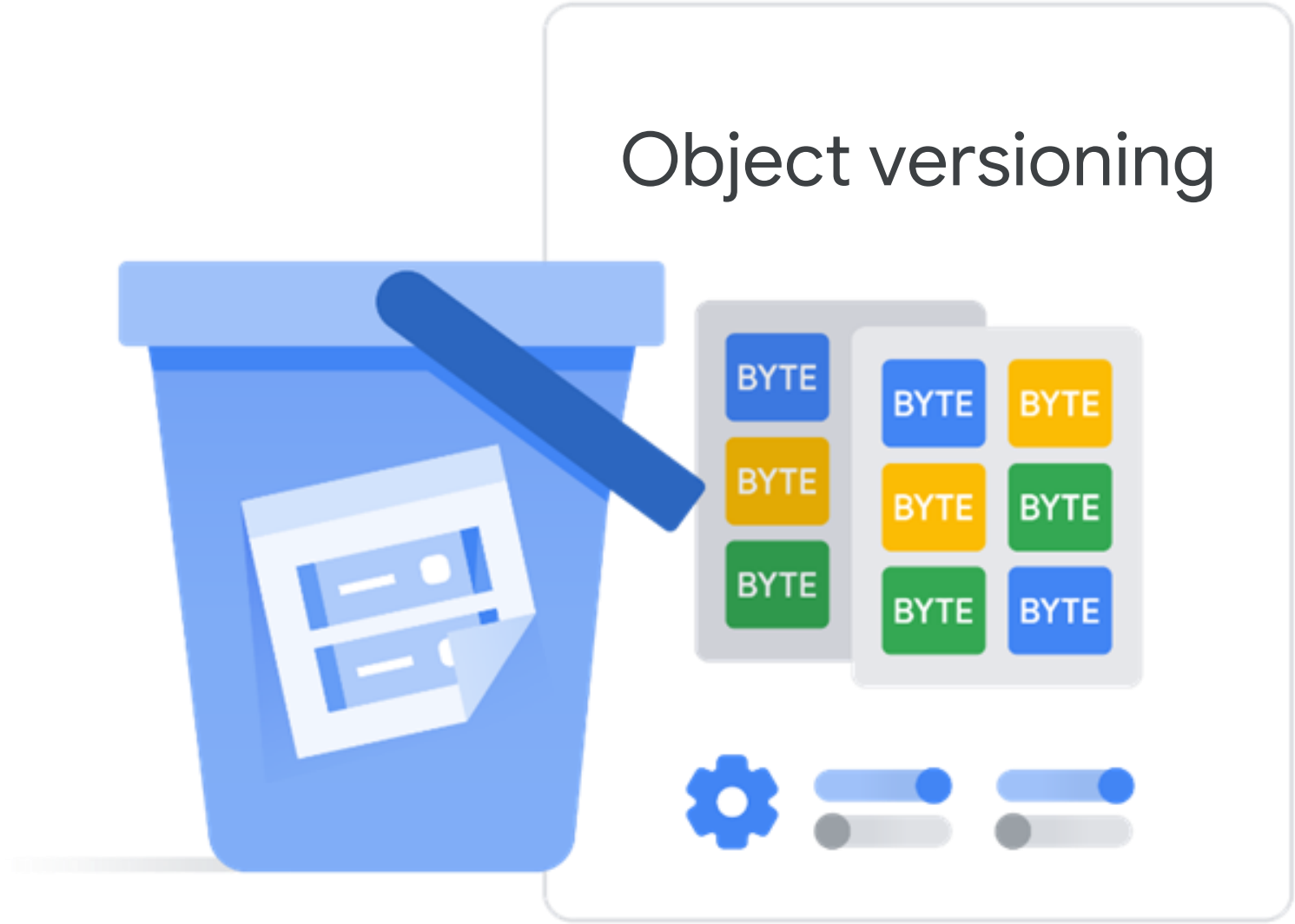


Cannot directly edit



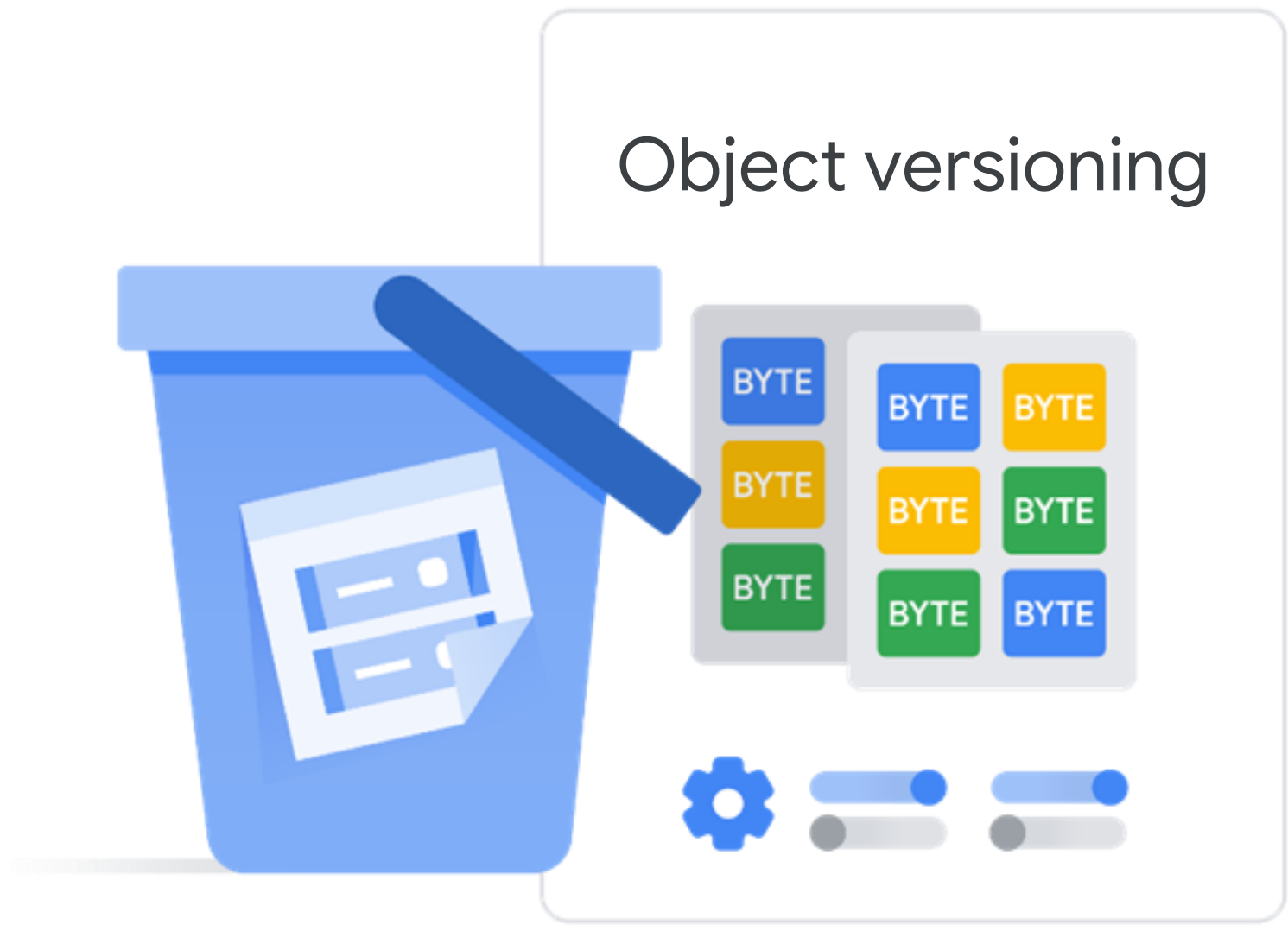
Create new version

Object versioning keeps a record of modifications



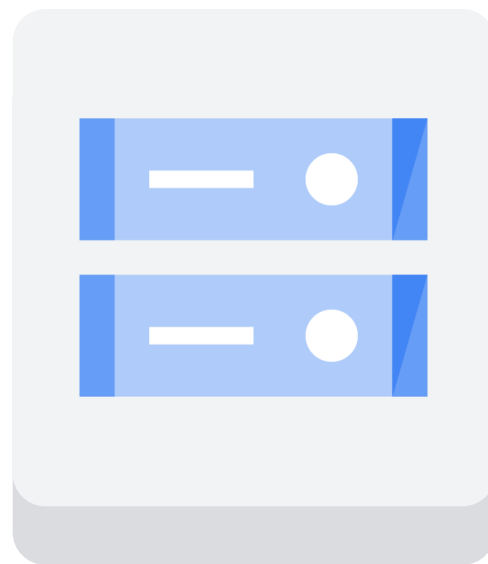
History	V1	V2	V3
Object 01	☒	☐	☐
Object 02	☒	☒	☒
Object 03	☒	☒	☐

Object versioning allows reversion to an older state



History	V1	V2	V3
Object 01	✓		
Object 02	✓	✓	✓
Object 03	✗	✓	

Integration with other Google Cloud products



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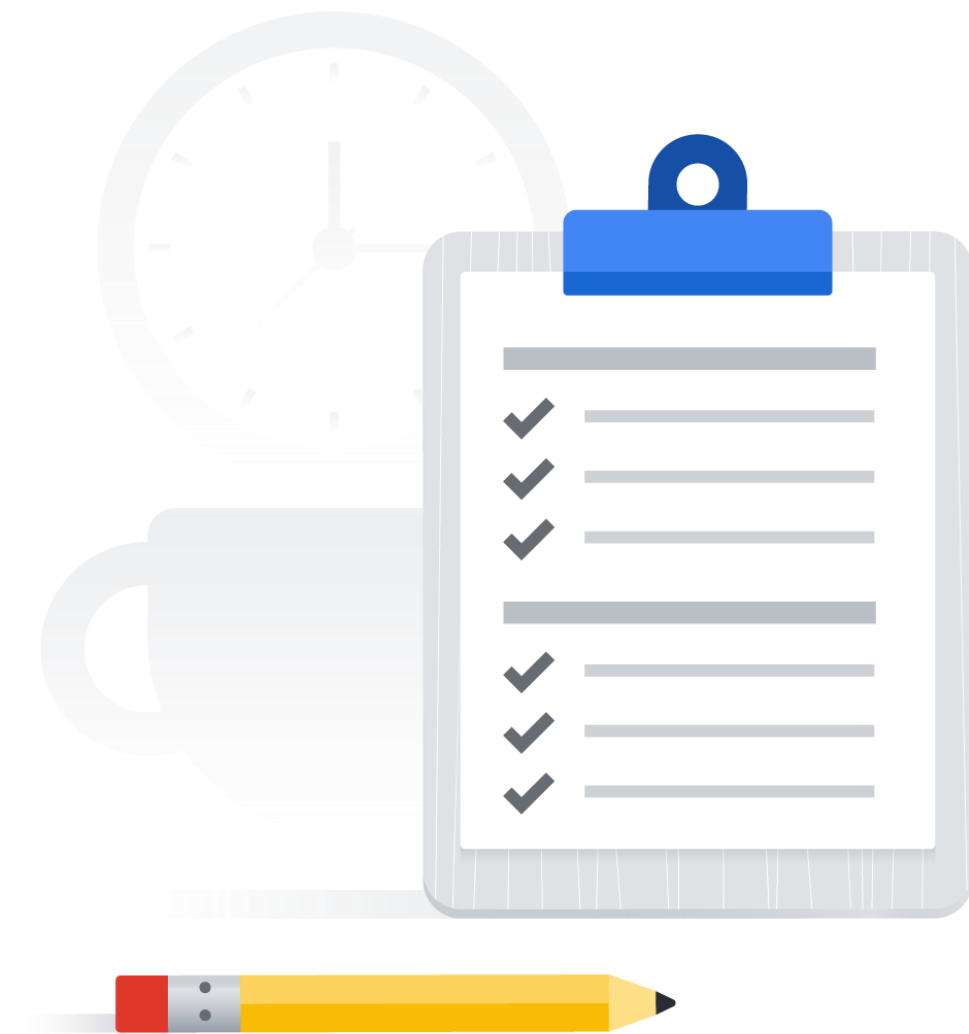
Agenda

Cloud Storage: Qwik Start - CLI/SDK

 Hands-on lab

During this lab, you'll:

1. Upload an object to a bucket.
2. Download an object from a bucket.
3. Copy an object to a folder in the bucket.
4. List contents of a bucket or folder.
5. List details for an object.
6. Make an object publicly accessible.



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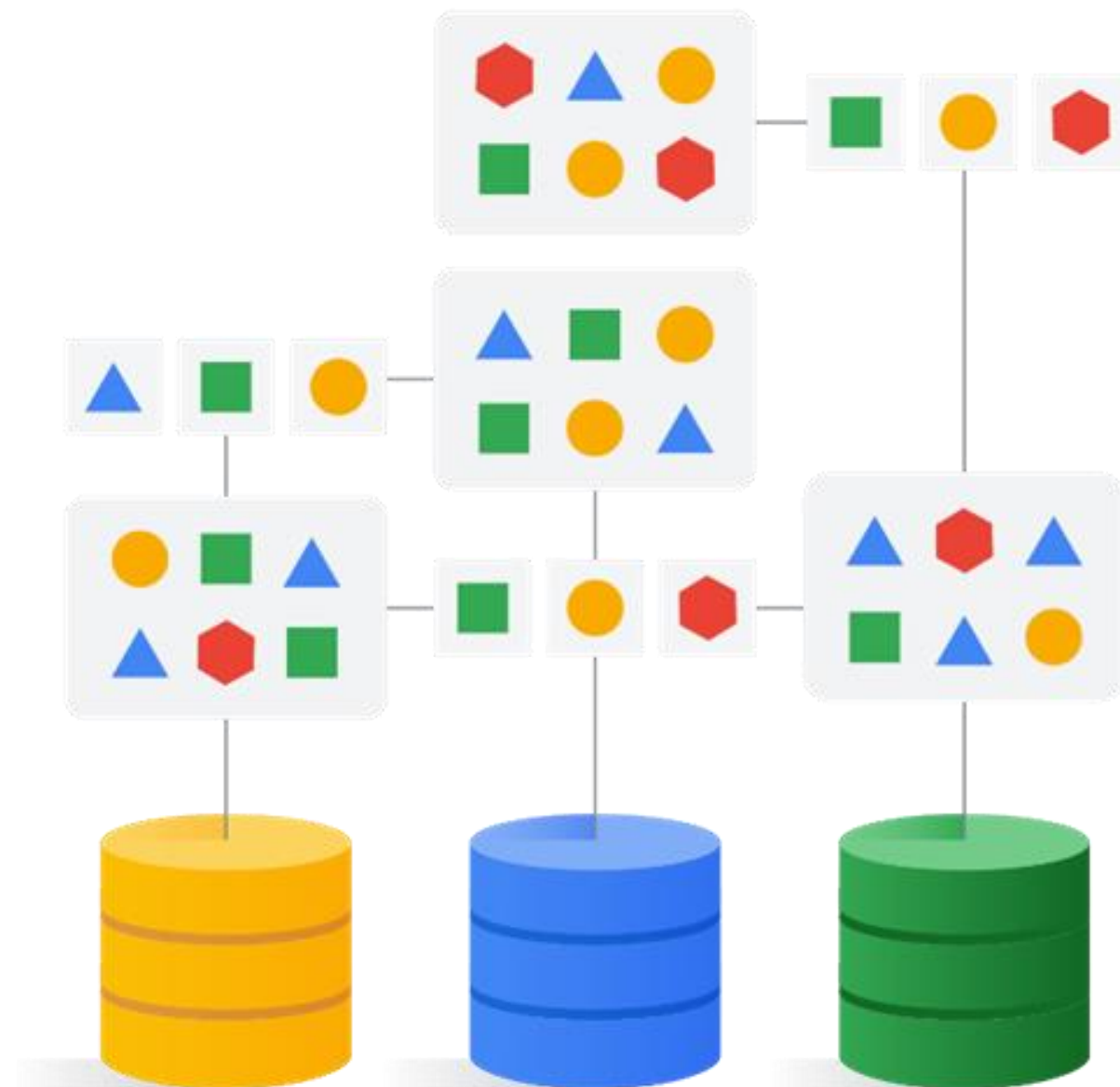
What is a database?

A collection of information organized so that it can easily be accessed and managed.

Computer applications run databases to get a fast answer to questions.



Relational databases are the most common



Relational database management systems

= RDBMS

= relational databases

= SQL databases

Suitable use cases:

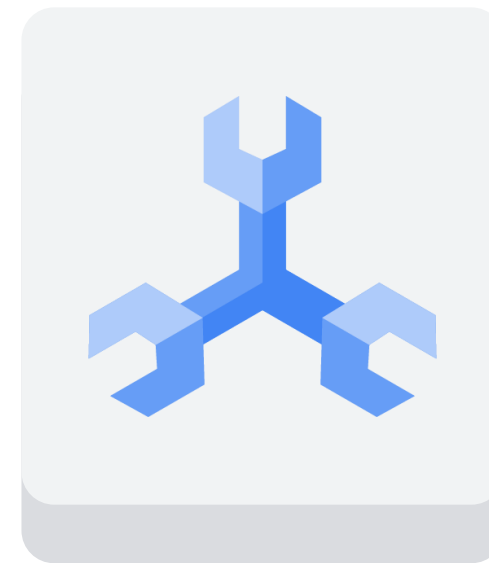
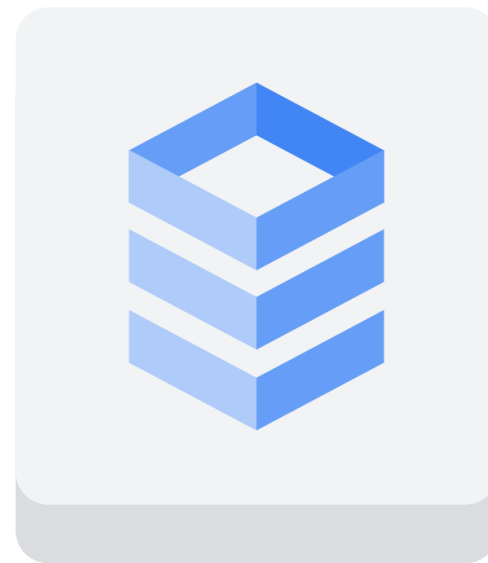
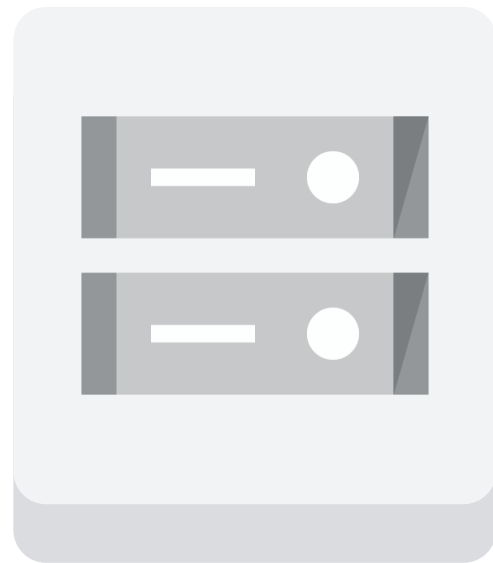
- Have a well-structured data model.
- Need transactions.
- Ability to join data across tables to retrieve complex data combinations.

Cloud SQL and Cloud Spanner

Relational
databases

Cloud SQL

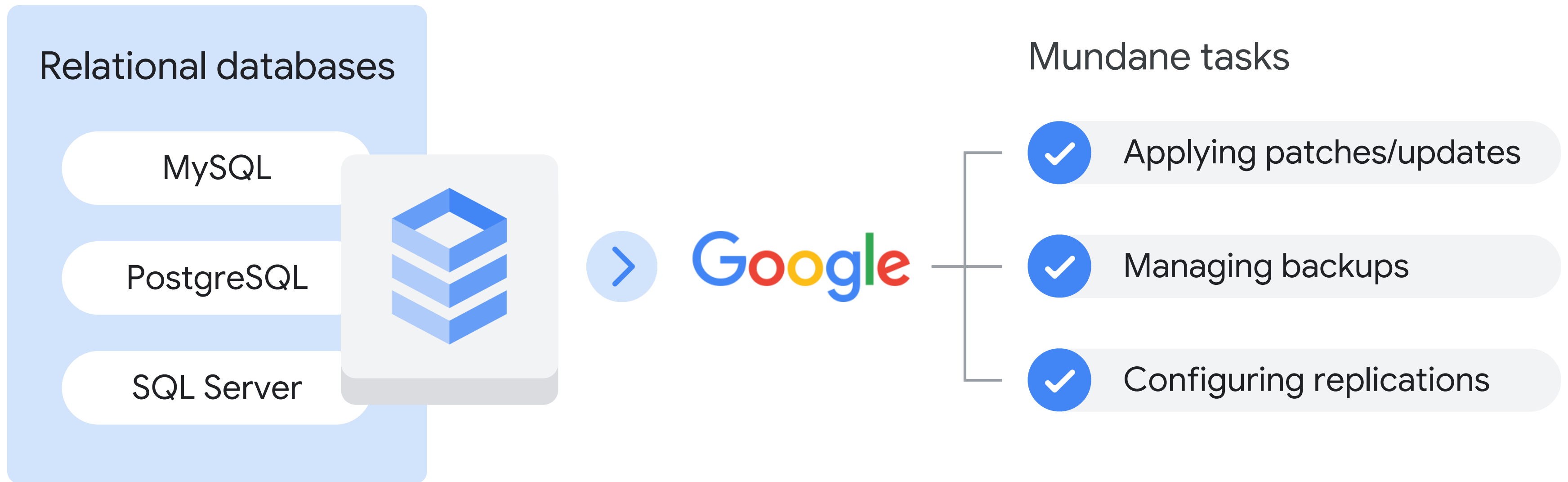
Cloud Spanner



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Cloud SQL lets you focus on building great applications



There are many reasons to use Cloud SQL

Doesn't require any software installation or maintenance

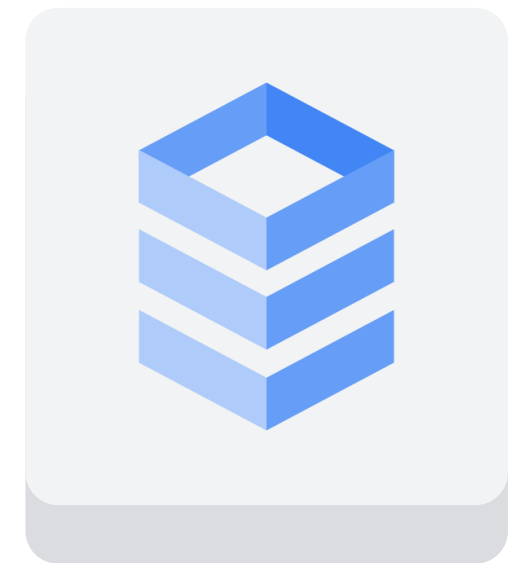
Can scale up to 96 processor cores, 624 GB of RAM, and 64 TB of storage

Supports automatic replication scenarios

Supports managed backups – The cost of an instance covers 7 backups

Encrypts customer data when on Google's internal networks and when stored in database tables, temporary files, and backups

Includes a network firewall



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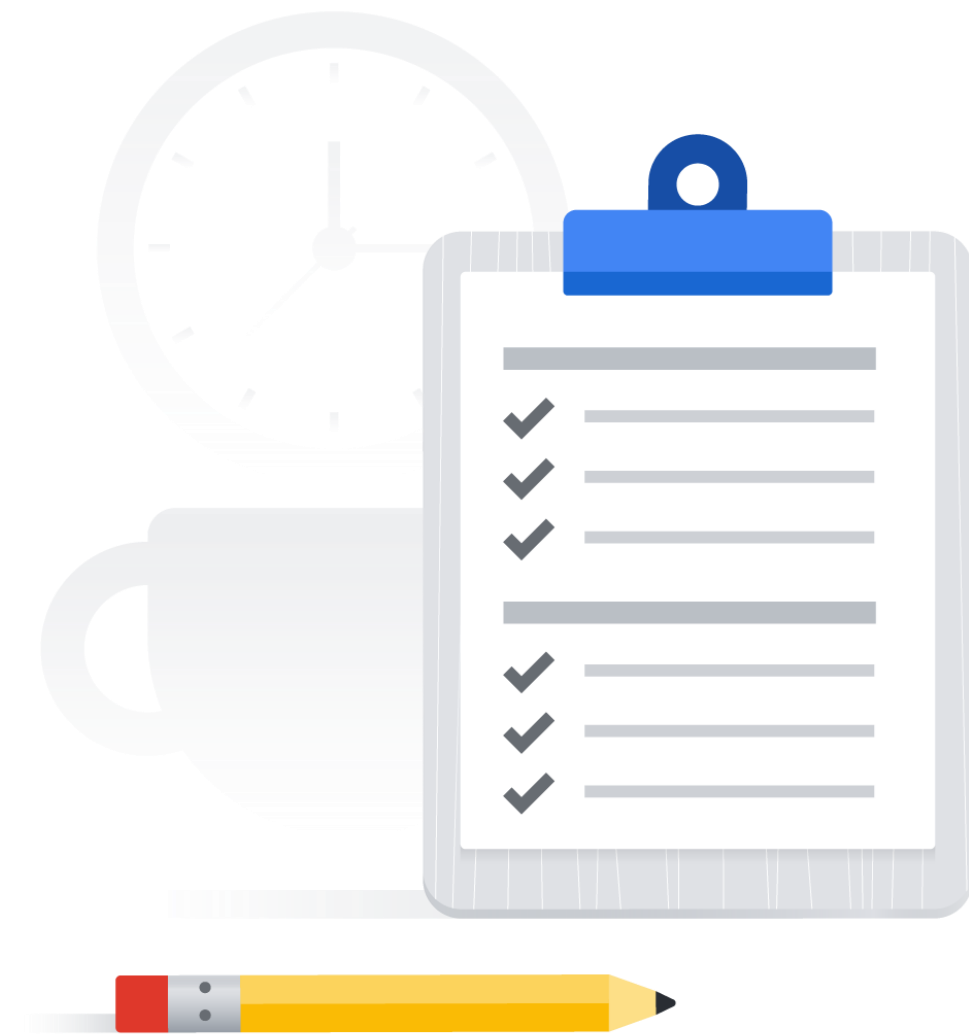
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Cloud SQL for MySQL: Qwik Start

 Hands-on lab

During this lab, you'll:

1. Create a Cloud SQL instance.
2. Connect to an instance using the mysql client in the Cloud Shell.

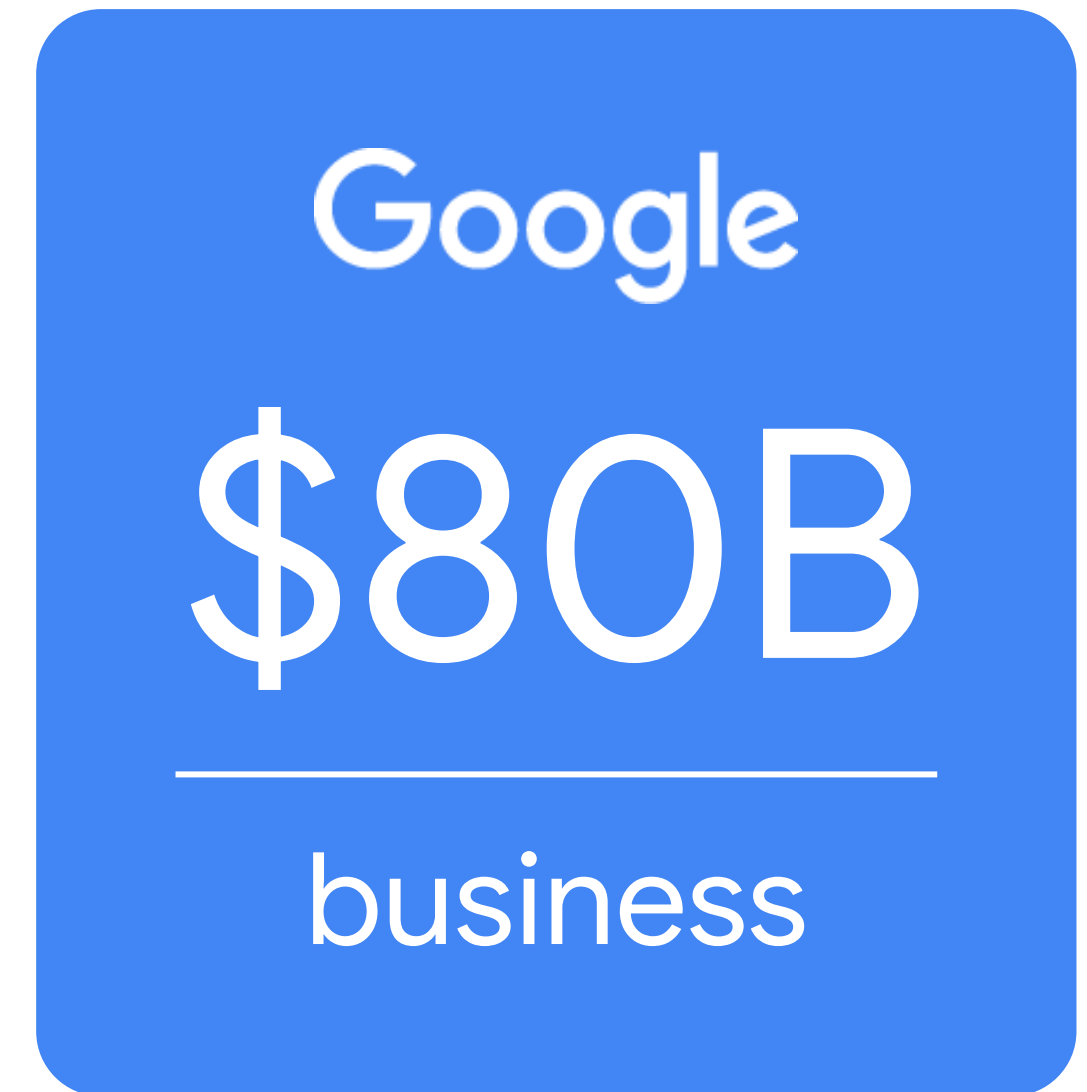
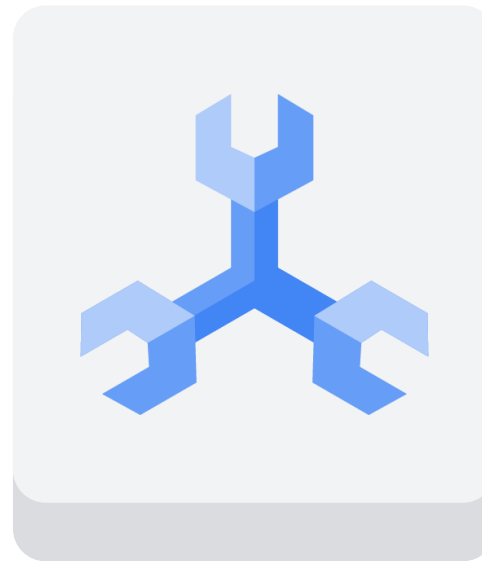


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Cloud Spanner is a fully managed relational database

- ✓ Scales horizontally
- ✓ Strongly consistent
- ✓ Speaks SQL



Applications that Cloud Spanner is especially suited for

Applications that require:

SQL relational database management system with joins and secondary indexes

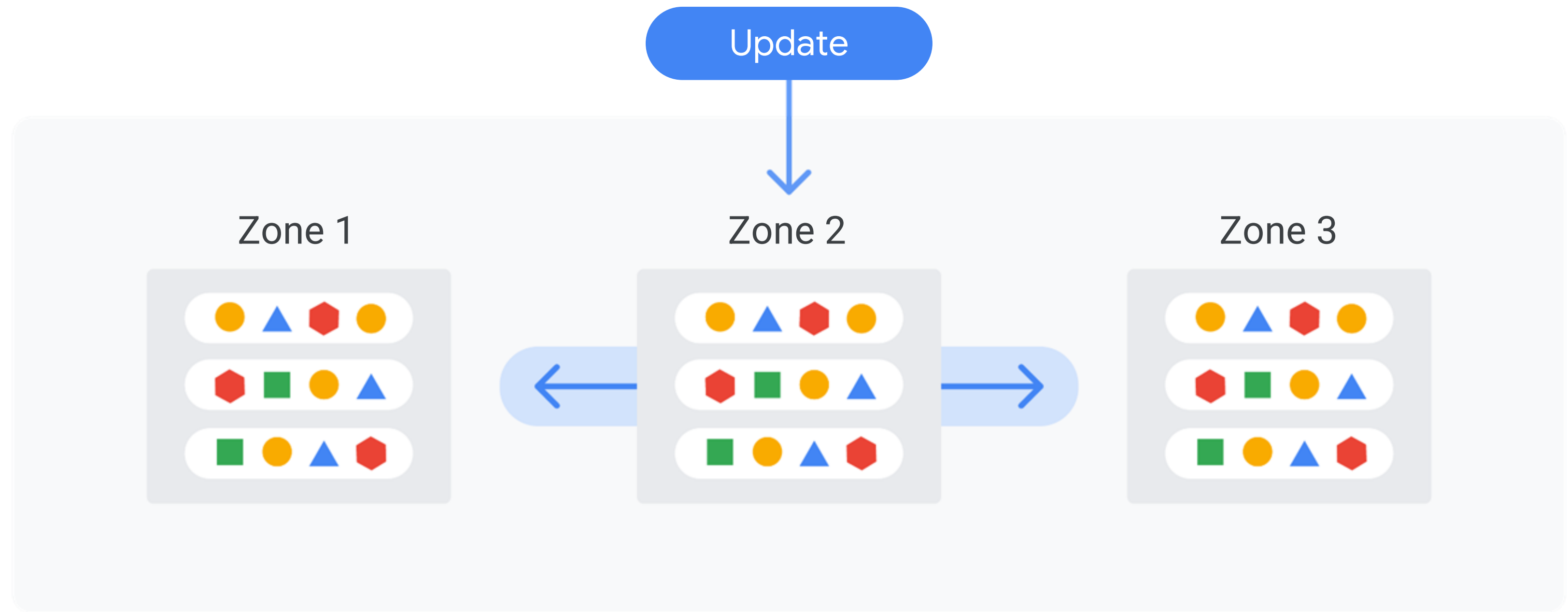
Built-in high availability

Strong global consistency

And high numbers of input/output operations per second. We're talking tens of thousands of reads/writes per second or more



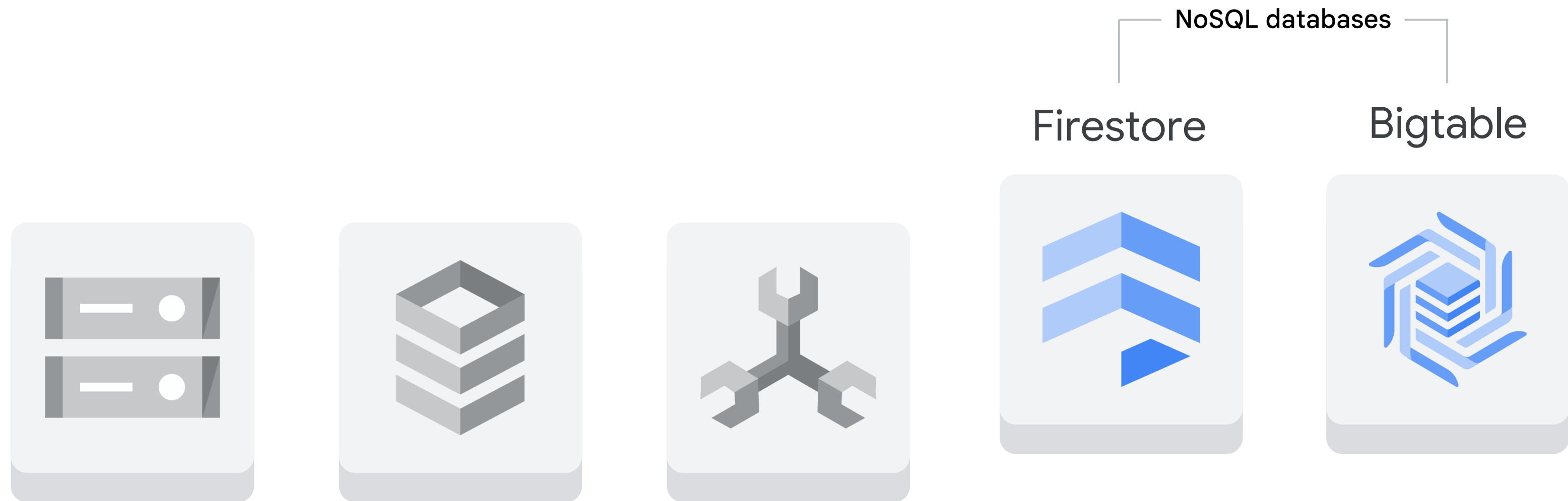
How does Cloud Spanner work?



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NoSQL-based managed services

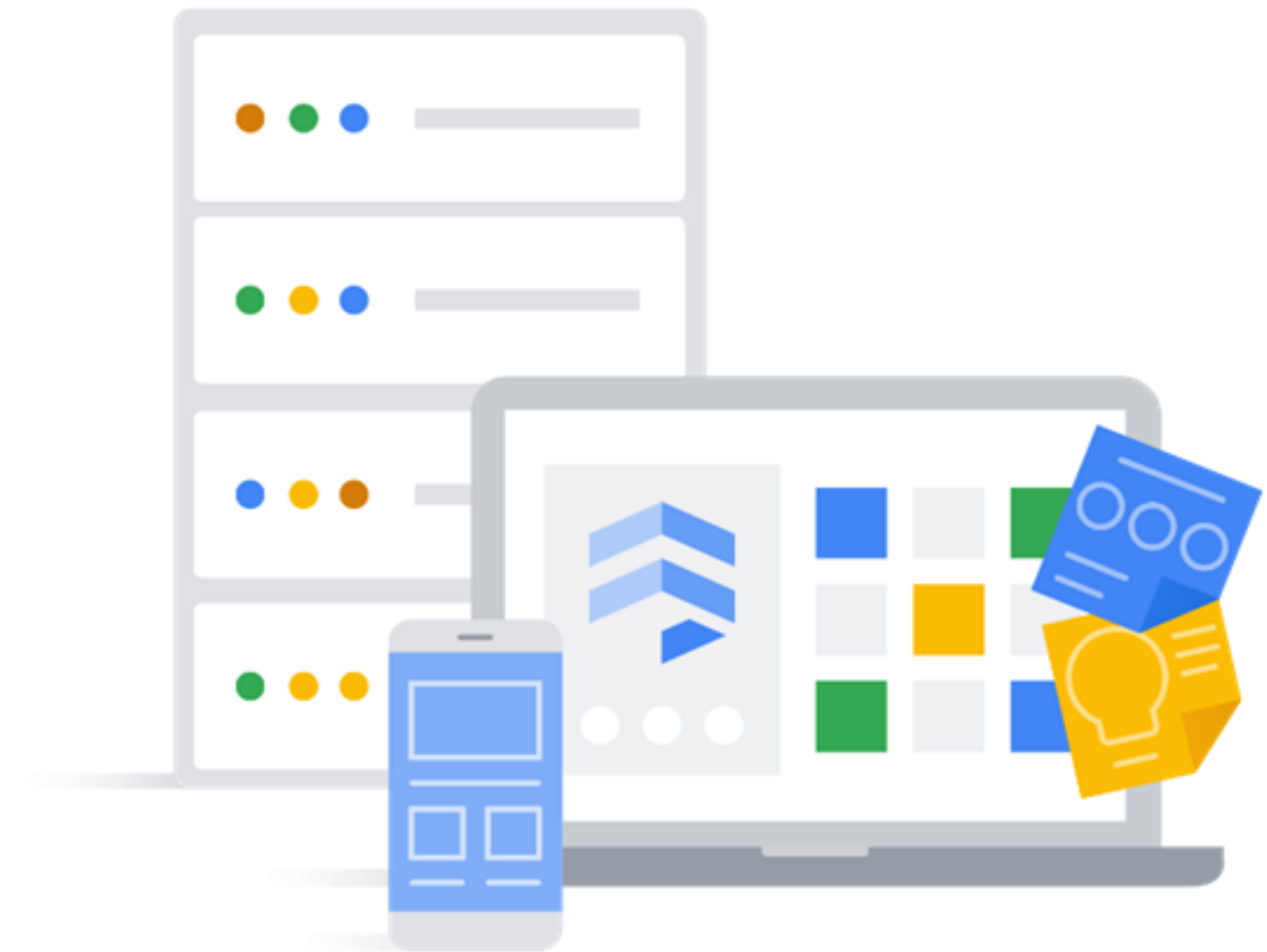


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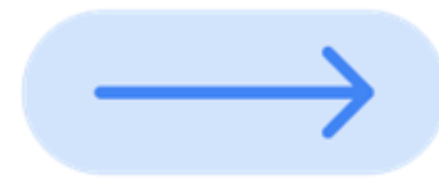
Firestore is ideal for mobile and web development

- ✓ Flexible
- ✓ Scales horizontally
- ✓ NoSQL cloud database

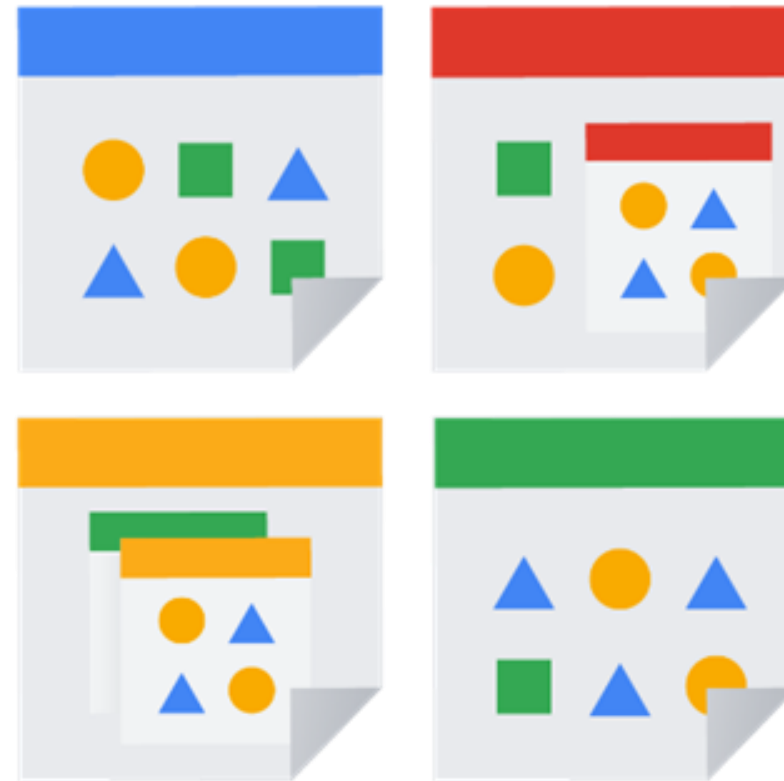


Firestore stores data as “documents”

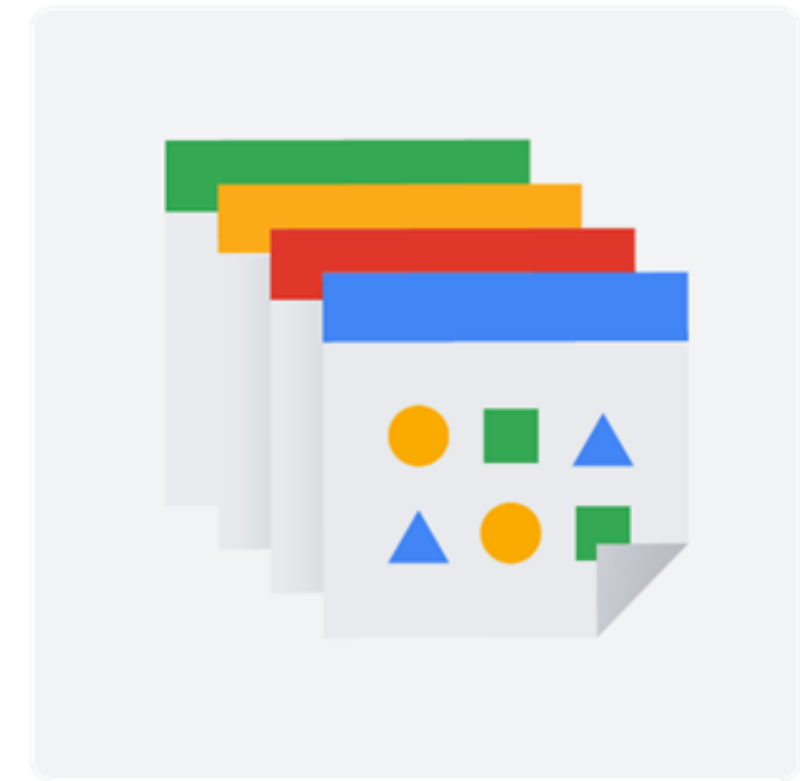
Incoming data



Documents

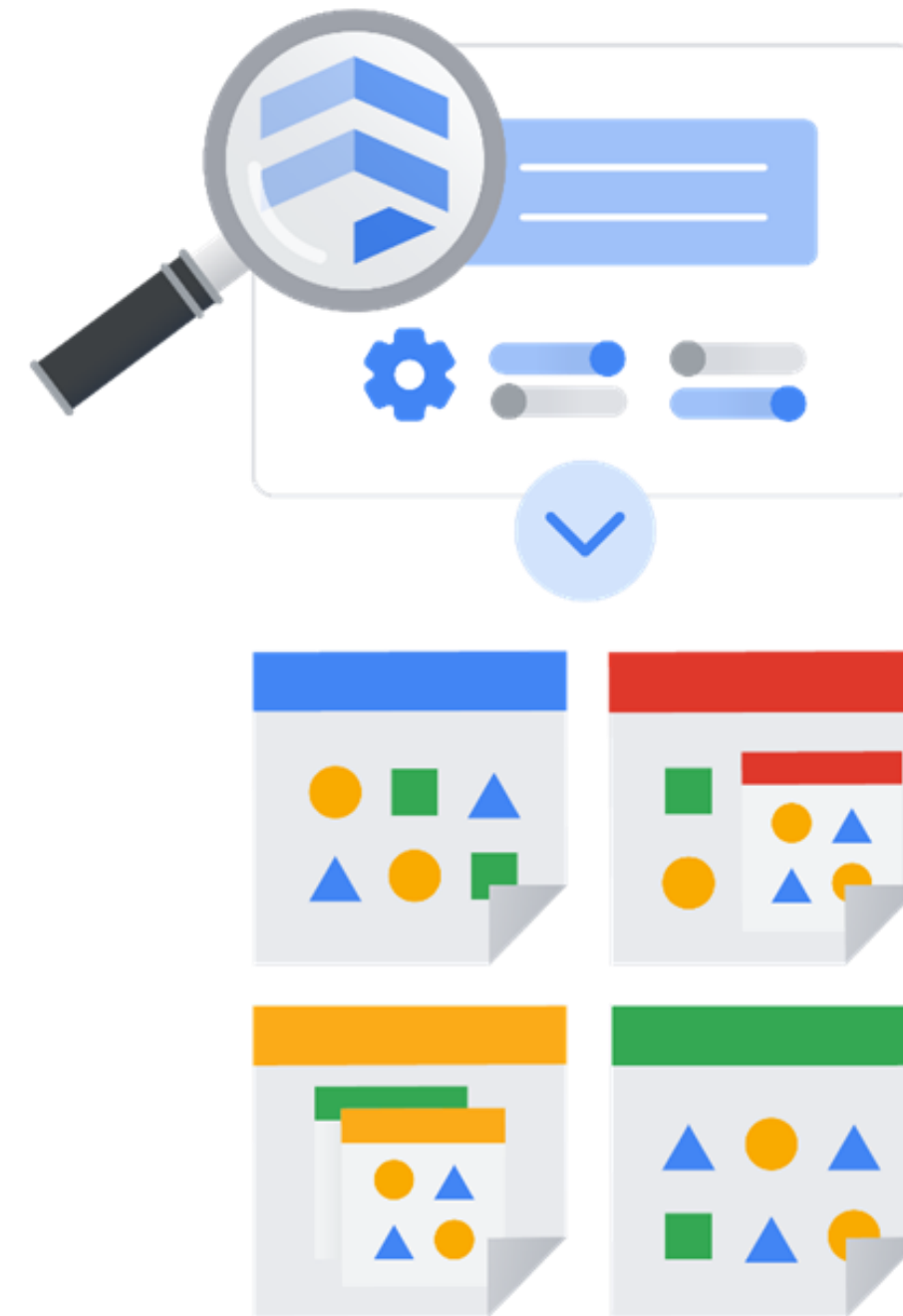


Collection

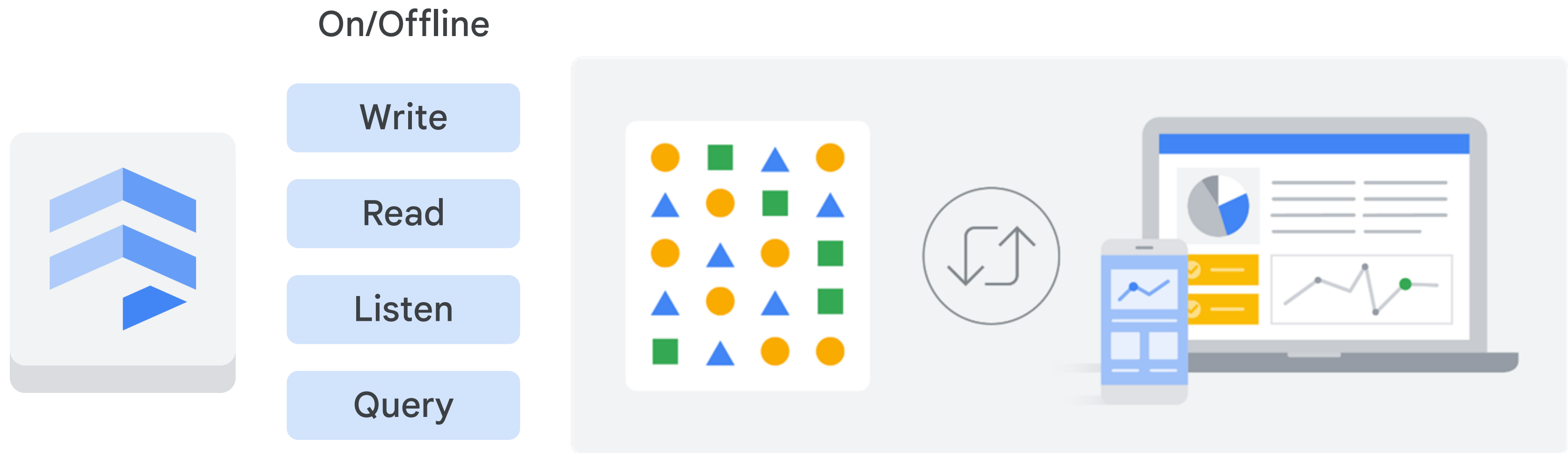


Firestore NoSQL queries connect apps with data

- ✓ Retrieve individual documents
- ✓ Retrieve all the documents in a collection
- ✓ Can include multiple, chained filters
- ✓ Can combine filtering and sorting options
- ✓ Indexed by default



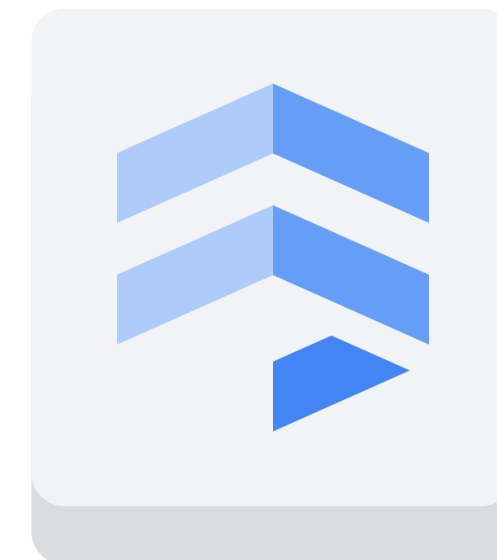
Firestore uses online and offline data synchronization



Firestore leverages Google Cloud's infrastructure

- ✓ Automatic multi-region data replication
- ✓ Strong consistency guarantees
- ✓ Atomic batch operations
- ✓ Real transaction support

Google Cloud



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Bigtable



Bigtable

Google's NoSQL big data database service

- ✓ Handle massive workloads
- ✓ Consistent low latency
- ✓ High throughput
- ✓ Great choice for:
- ✓ Operational applications
- ✓ Analytical applications

Why customers often choose Bigtable

Work with more than 1TB of semi-structured or structured data

Data is fast with high throughput, or it's rapidly changing

Work with NoSQL data

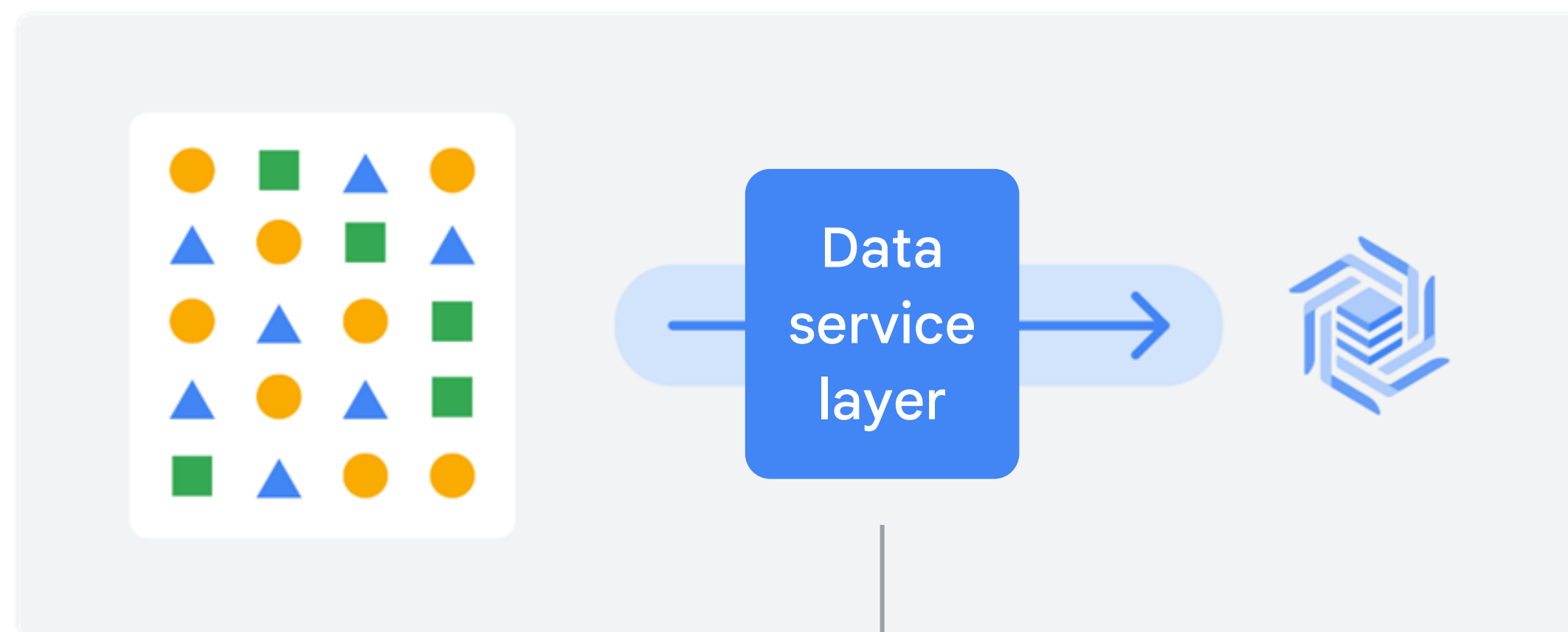
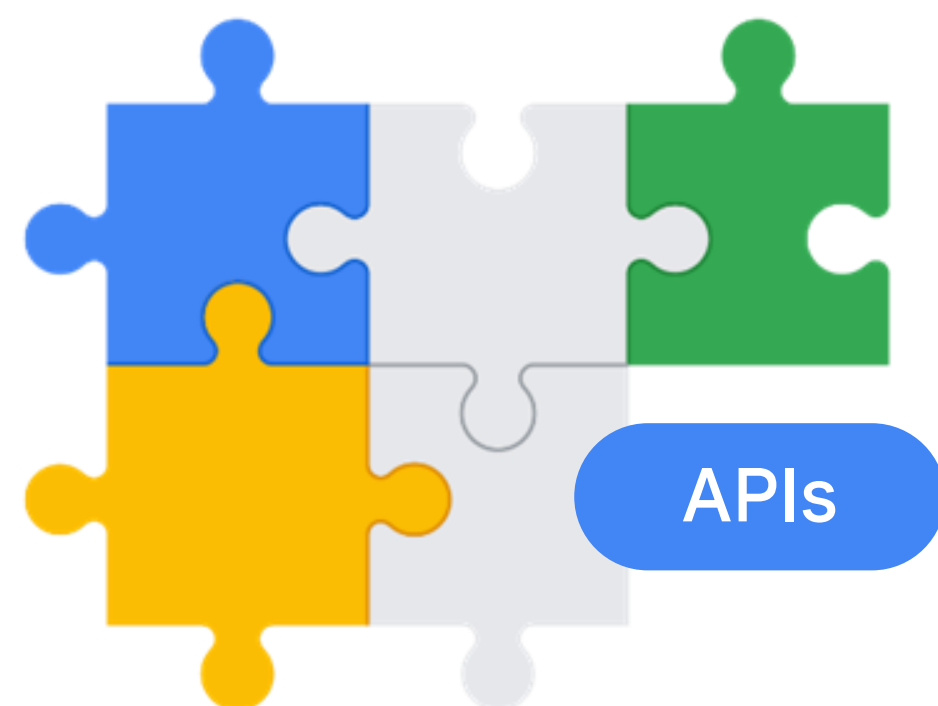
Data is a time-series or has natural semantic ordering

Work with big data, running asynchronous batch or synchronous real-time processing on the data

Run machine learning algorithms on the data



Using APIs to read and write data



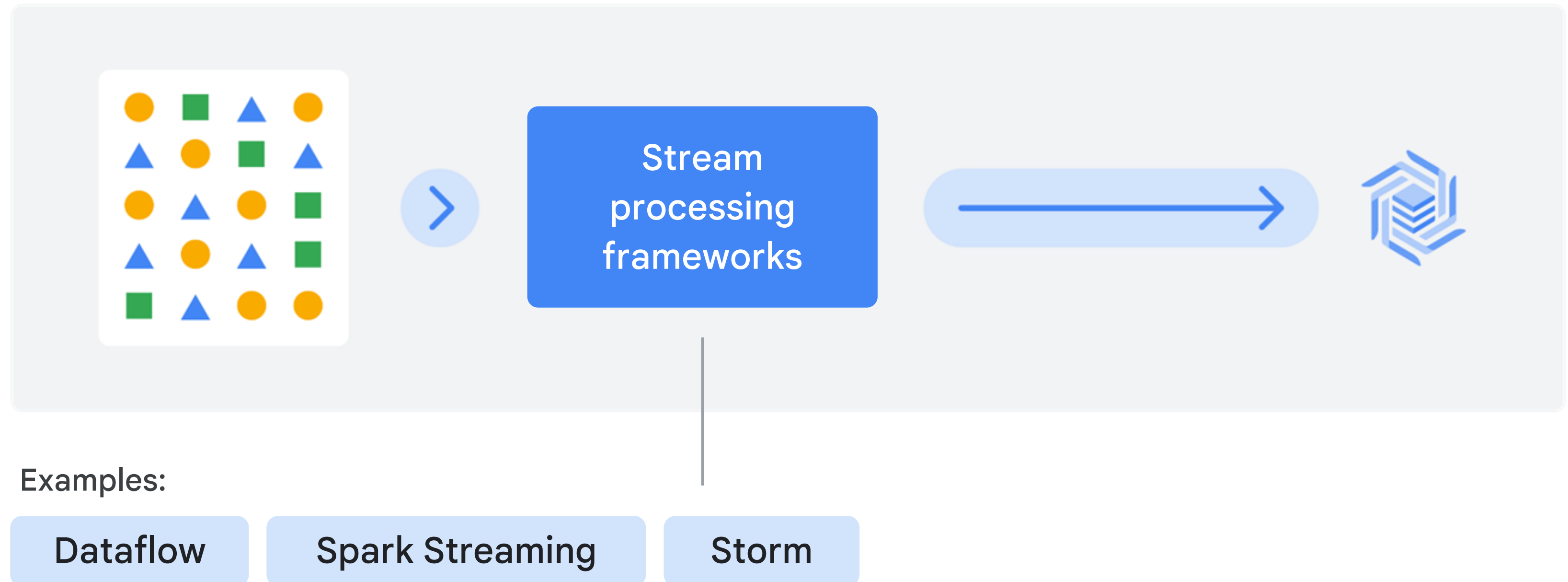
Examples:

Managed VMs

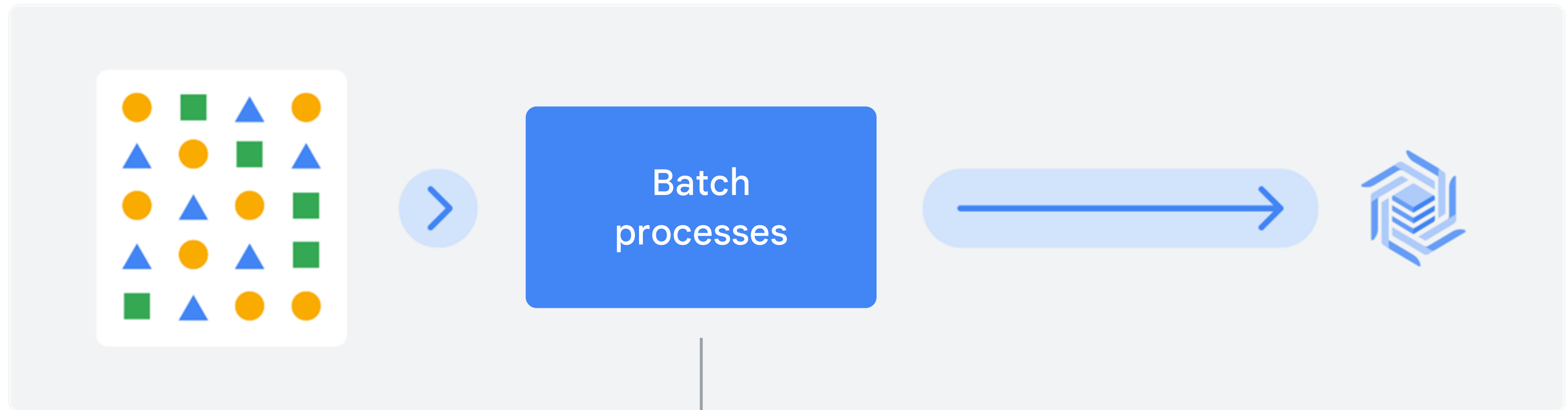
HBase REST Server

Java Server

Streaming data into Bigtable



Using batch processes to read and write data



Examples:

Hadoop MapReduce

Dataflow

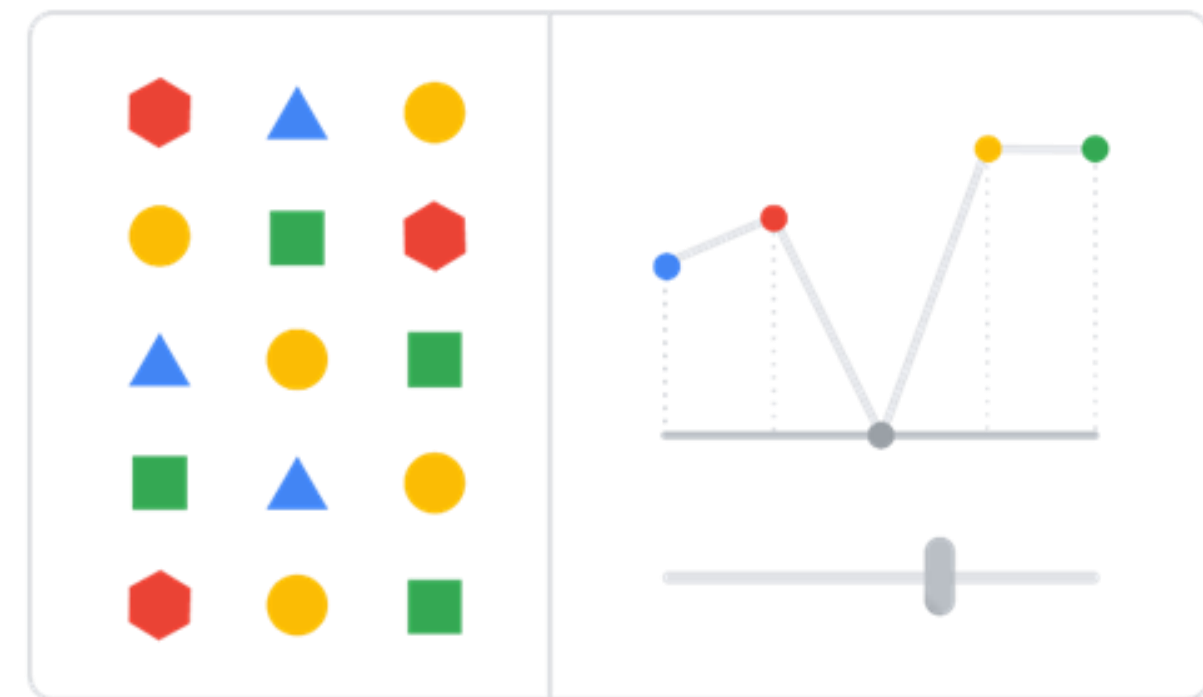
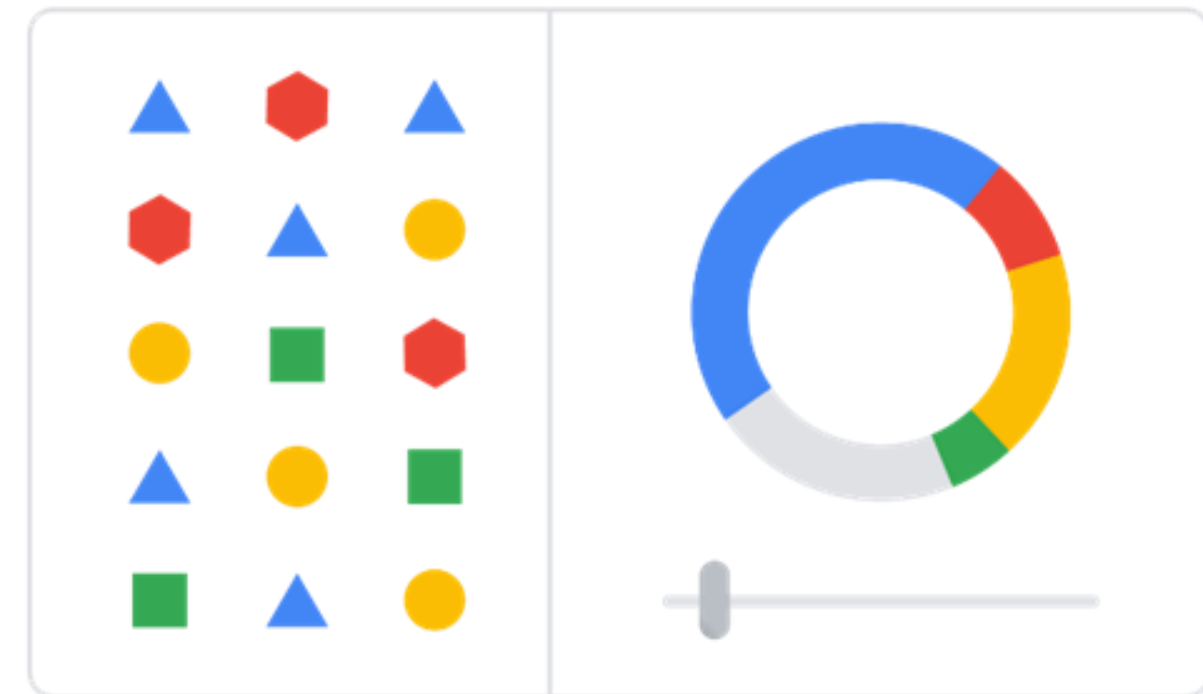
Spark

BigQuery and Bigtable are different products with different uses



BigQuery

On the edge between data storage and data processing



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Knowledge check

Which storage service is best suited to unstructured data?

- A. Cloud Storage
- B. Cloud Spanner
- C. Firestore
- D. Bigtable



Knowledge check

Which storage service is best suited to unstructured data?

A. Cloud Storage

B. Cloud Spanner

C. Firestore

D. Bigtable



Knowledge check

You are looking for an unstructured storage solution for archiving files that might never be accessed again. Which Cloud Storage class is the best option?

- A. Standard storage
- B. Nearline storage
- C. Coldline storage
- D. Archive storage



Knowledge check

You are looking for an unstructured storage solution for archiving files that might never be accessed again. Which Cloud Storage class is the best option?

- A. Standard storage
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Knowledge check

Google Cloud offers two managed relational database services.
What are they?

- A. Cloud Bigtable
- B. Cloud Spanner
- C. Firestore
- D. Cloud SQL



Knowledge check

Google Cloud offers two managed relational database services.
What are they?

A. Cloud Bigtable

B. Cloud Spanner

C. Firestore

D. Cloud SQL



Knowledge check

Which storage solution is a petabyte scale, NoSQL database?

- A. Bigtable
- B. Cloud Spanner
- C. Firestore
- D. Cloud SQL



Knowledge check

Which storage solution is a petabyte scale, NoSQL database?

A. Bigtable

B. Cloud Spanner

C. Firestore

D. Cloud SQL



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Summary

- ✓ Explored the different storage options available in Google Cloud.
- ✓ Learned to differentiate between structured and unstructured storage in the cloud.
- ✓ Examined how you can use Cloud Storage for unstructured data storage.
- ✓ Explored the use case for relational vs NoSQL storage options, and identify the options available with Google Cloud.



